

2. potęgi 1

Oblicz:

a) $\left(\frac{3}{\sqrt{2}}\right)^{-4}$ b) $\left(-\frac{\sqrt{5}}{10}\right)^{-3}$ c) $\left((-3\sqrt{2})^{-1}\right)^{-4}$

a) $\left(\frac{3}{\sqrt{2}}\right)^{-4}$ $a^{-n} = \frac{1}{a^n}$ $\left(\frac{a}{b}\right)^r = \frac{a^r}{b^r}$

$$\left(\frac{3}{\sqrt{2}}\right)^{-4} = \left(\frac{\sqrt{2}}{3}\right)^4 = \frac{4}{81}$$

odp: $\frac{4}{81}$.

b) $\left(-\frac{\sqrt{5}}{10}\right)^{-3} = \left(-\frac{10}{\sqrt{5}}\right)^3 = -\frac{1000}{5\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} =$
 $= -\frac{1000\sqrt{5}}{25} = -40\sqrt{5}$

odp: $-40\sqrt{5}$.

c) $\left((-3\sqrt{2})^{-1}\right)^{-4} = (-3\sqrt{2})^4 = 81 \cdot 4 = 324$

odp: 324.